OPERATING SYSTEM

ORANGE PRIOBLEM-2

NAME:- DEVARAJ M

SRN:- PES1UG24AM079

2ND YEAR AIML

“B” SECTION

```
@devrajm2005-debu ➜/workspaces/PES1UG24AM079-PES-VCS (main) $ make clean
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
@devrajm2005-debu ➜/workspaces/PES1UG24AM079-PES-VCS (main) $ make all
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
@devrajm2005-debu ➜/workspaces/PES1UG24AM079-PES-VCS (main) $ ./pes init
echo "Hello" > hello.txt
./pes add hello.txt
export PES_AUTHOR="Devaraj M <PES1UG24AM079>"
./pes commit -m "Initial commit"
./pes log
Initialized empty PES repository in .pes/
Committed: 1857066b7981... Initial commit
commit 1857066b7981d8e4073b3535fb0e32e54694f0e9c5e162dfbf53227e1e659f6f
Author: Devaraj M <PES1UG24AM079>
Date:   1776765049

    Initial commit
```

```
● @devrajm2005-debu ➜/workspaces/PES1UG24AM079-PES-VCS (main) $  ./pes log
find .pes -type f | sort
cat .pes/refs/heads/main
cat .pes/HEAD
commit 1857066b7981d8e4073b3535fb0e32e54694f0e9c5e162dfbf53227e1e659f6f
Author: Devaraj M <PES1UG24AM079>
Date:   1776765049

    Initial commit

.pes/HEAD
.pes/index
.pes/objects/18/57066b7981d8e4073b3535fb0e32e54694f0e9c5e162dfbf53227e1e659f6f
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/refs/heads/main
1857066b7981d8e4073b3535fb0e32e54694f0e9c5e162dfbf53227e1e659f6f
ref: refs/heads/main
● @devrajm2005-debu ➜/workspaces/PES1UG24AM079-PES-VCS (main) $ echo "world" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"

./pes log
Committed: 345a6a2c4abf... Add world
Committed: f8d5d5ef81d5... Add farewell
commit f8d5d5ef81d5cd754b858015236625e9bfe0c358ffc8eb4f77577f531c195a48
Author: Devaraj M <PES1UG24AM079>
Date:   1776765333

    Add farewell

commit 345a6a2c4abf85dfc3c2b2b110a514d5a5d663e2e70bb2f08f2bb1228385403f
Author: Devaraj M <PES1UG24AM079>
Date:   1776765333

    Add worldell
```

```
@devrajm2005-debu ➜/workspaces/PES1UG24AM079-PES-VCS (main) $ echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Goodbye" > bye.txt …

    Add worldell

commit 1857066b7981d8e4073b3535fb0e32e54694f0e9c5e162dfbf53227e1e659f6f
Author: Devaraj M <PES1UG24AM079>
Date:   1776765049

    Initial committial comQ

● @devrajm2005-debu ➜/workspaces/PES1UG24AM079-PES-VCS (main) $ make test_objects
./test_objects

make test_tree
./test_tree
make: 'test_objects' is up to date.
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
make: 'test_tree' is up to date.
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
● @devrajm2005-debu ➜/workspaces/PES1UG24AM079-PES-VCS (main) $ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists
```

```
@devrajm2005-debu ➜/workspaces/PES1UG24AM079-PES-VCS (main) $ make test-integration
Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 4596e002afeb... Initial commit

Log after first commit:
commit 4596e002afeb289771e738e532be099f2d9852617d51a8d1be33ae8bf00e4420
Author: Devaraj M <PES1UG24AM079>
Date:   1776765408

    Initial commit


--- Second Commit ---
Committed: 7ef53b9a5856... Update file.txt

--- Third Commit ---
Committed: fcd66f8fb20c... Add farewell

--- Full History ---
commit fcd66f8fb20c60e80e2ecda246216c0aa122f521aa76e2f8769f59dfb5c31886
Author: Devaraj M <PES1UG24AM079>
Date:   1776765408

    Add farewell

commit 7ef53b9a5856a50ab214fb913e22b81c9fbeb6bba018628146595b738befdd4d
Author: Devaraj M <PES1UG24AM079>
Date:   1776765408

    Update file.txt

commit 4596e002afeb289771e738e532be099f2d9852617d51a8d1be33ae8bf00e4420
Author: Devaraj M <PES1UG24AM079>
Date:   1776765408

    Initial committial comQ
```

```
@devrajm2005-debu ➜/workspaces/PES1UG24AM079-PES-VCS (main) $
@IAmKai1 ➜/workspaces/PES1UG24AM084-pes-vcs1 (main) $ ./test_objects
./test_tree
./pes log
find .pes -type f | sort
cat .pes/refs/heads/main
cat .pes/HEAD
bash test_sequence.sh
bash: syntax error near unexpected token `('
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
No commits yet.
cat: .pes/refs/heads/main: No such file or directory
cat: .pes/HEAD: No such file or directory
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 6360c6963735... Initial commit

Log after first commit:
```

```
@devrajm2005-debu ➜/workspaces/PES1UG24AM079-PES-VCS (main) $
@IAmKai1 ➜/workspaces/PES1UG24AM084-pes-vcs1 (main) $ ./test_objects
./test_tree
./pes log
find .pes -type f | sort …

    Add farewell

commit ce3fc8a92325a54b996e14447847aa8e5f225a6cd10ccf1b34a1743f33bee6cb
Author: Devaraj M <PES1UG24AM079>
Date:   1776765477

    Update file.txt

commit 6360c6963735fe9ab6eece42ba0237a4e3d7ee0a842993deb97c4c9be4a29b4d
Author: Devaraj M <PES1UG24AM079>
Date:   1776765477

    Initial committial comQ


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
0a7cddb08efab933b22b3887ecd33408dc3a6b2d96612051fb316eb899433872

--- Object Store ---
Objects created:
10
.pes/objects/0a/7cddb08efab933b22b3887ecd33408dc3a6b2d96612051fb316eb899433872
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/63/60c6963735fe9ab6eece42ba0237a4e3d7ee0a842993deb97c4c9be4a29b4d
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/ce/3fc8a92325a54b996e14447847aa8e5f225a6cd10ccf1b34a1743f33bee6cb
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
@devrajm2005-debu ➜/workspaces/PES1UG24AM079-PES-VCS (main) $
```